

B

[RUBRIC
B]ASSESSMENT
INSTRUMENT

Mission Decision Quality — AI Judge Logic

รูบริก B · คุณภาพการตัดสินใจในการกิจ (ตรรกะ AI Judge)

PURPOSE · วัตถุประสงค์

Make the AI Judges' scoring logic **transparent and auditable** — for teachers, learners, and the competition judges. Every decision a learner submits inside a mission is scored against the criteria in this rubric. The platform's feedback prompts and Insight-Token awards are deterministic functions of these scores.

OBJECTIVES · เป้าหมายเฉพาะ

- Specify the exact dimensions on which AI Judges score each of the five mission stages (BRIEF → PROBE → DECIDE → ACT → DEBRIEF).
- Anchor every AI prompt to descriptors a teacher could audit by hand on a random-sample basis.
- Document the deliberate confidence-trap at the DECIDE stage and why it is pedagogically defensible.
- Define how stage scores compose into the mission pass-threshold that awards an SDG Keystone (idempotent — keystones cannot be double-counted).
- Specify the formative-feedback contract: every return includes a specific actionable `growthEdge`; a null fails validation and the call retries — the journey never blocks.

APPLIES TO · ขอบเขตการใช้

Each decision logged during a mission. Six regional missions (EEC fringe health · Mae Sot education · Khon Kaen water · Bangkok cities · Chiang Mai climate · Andaman marine) all run the same 5-stage arc. The AI Judge scores per stage; the engine composes the mission pass-threshold from those scores.

SCORING SCALE · มาตรการให้คะแนน

Each criterion 1 · 3 · 5. Insight Tokens awarded -10 ... +25 from the integrated score (tokens are a process reward — they never buy a Keystone). Pass threshold: engine composite ≥ 0.60 OR net-token bar ≥ 8 ♦. Clearing the bar awards the SDG Keystone for that region (idempotent).

USED BY · ผู้ใช้รูบริก

AI Judges (Tier 1 of the 4-tier hybrid judging system, formative, every mission, every region). Teachers audit randomly-sampled scores against this rubric to calibrate the model. *The capstone (Voice for Change) is excluded by design* — it belongs to Rubric A (teacher), preserving the AI-TPACK boundary.

PEDAGOGICAL ANCHORS · กรอบทฤษฎีที่ใช้

AI-TPACK (boundary) · PICRAT · Interactive-Amplifies · Bloom's Revised Taxonomy
(Remember → Metacognition) · Hattie Visible Learning (formative feedback) · Vygotsky
ZPD (the confidence-trap forces productive struggle)

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STAGE 01 • BRIEF — ABSORB THE SITUATION

BLOOM: REMEMBER / UNDERSTAND

CRITERION	1 • EMERGING	3 • DEVELOPING	5 • DISTINGUISHED
Factual grounding	Multiple errors of fact	Mostly accurate, minor slips	All claims traceable to the dossier (at the learner's tier)
Vocabulary recognition	Misuses key terms	Uses key terms with occasional imprecision	Uses key terms precisely; differentiates near-synonyms
Situation grasp	Misses the core tension	States the tension	Names the tension <i>and</i> whose interests it pits

STAGE 02 • PROBE — QUESTION THE SOURCES

BLOOM: ANALYZE

CRITERION	1 • EMERGING	3 • DEVELOPING	5 • DISTINGUISHED
Source provenance	Trusts the most familiar / loudest source	Weights one provenance signal	Weights methodology, recency, and conflict of interest together
Comprehension depth	Surface restatement	Captures one layer	Captures multiple layers and the contested claim
Stakeholder reading	Conflates stakeholders	Maps each to a stated concern	Distinguishes stated vs. likely hidden concerns

STAGE 03 • DECIDE — CHOOSE UNDER A REAL TRADE-OFF

BLOOM: EVALUATE • HIGHEST-STAKES SCORING

CRITERION	1 • EMERGING	3 • DEVELOPING	5 • DISTINGUISHED
Strategic coherence	Choice contradicts stated values	Aligns but generic	Specific, defensible, traceable to scenario evidence
Consequence awareness	Ignores second-order effects	Names one consequence	Anticipates the trade-off the counter-position will raise
Confidence calibration	High confidence on weak reasoning (<i>deliberate trap</i>)	Confidence loosely matched to evidence	Confidence honestly tracks evidence strength

TOKEN BAND

The arc's highest-stakes scoring — -10 (high-confidence trap) ... +25 (calibrated, defensible). The confidence-trap is deliberate: it makes overclaiming costly, modelling intellectual humility (Vygotsky ZPD-style productive struggle).

STAGE 04 • ACT — COMMUNICATE THE DECISION

BLOOM: APPLY

CRITERION	1 • EMERGING	3 • DEVELOPING	5 • DISTINGUISHED
Language quality	Errors compromise meaning	Generally clear, minor errors	Controlled register, audience-appropriate
Audience awareness	Generic "audience"	Names one audience-specific concern	Anticipates and addresses audience objections
Justification	One-sentence assertion	Reasoned but thin	Explicit, specific, names the trade-off

STAGE 05 • DEBRIEF — REFLECT ON CONSEQUENCE

BLOOM: METACOGNITION

CRITERION	1 • EMERGING	3 • DEVELOPING	5 • DISTINGUISHED
Counterargument handling	Dismisses or capitulates	Acknowledges, partially answers	Acknowledges, refutes, refines own position
Willingness to revise	None	Revision in form only	Genuinely incorporates the strongest counter
Evidence commitment	None recorded	Names evidence vaguely	Commits to specific evidence (carried to the Voice for Change)

CALIBRATION & OPERATIONAL NOTES

- **AI-Judge calibration:** during the pilot, random-sample ~5% of decisions and compare AI scores to a teacher's; drift greater than 0.5 score-point on any criterion triggers prompt revision.
- **growthEdge mandatory:** every AI return must include a specific actionable growthEdge . A null fails validation and the call is retried — the journey never blocks.
- **No dead-ends:** a learner below the bar may replay DECIDE/ACT, or a teacher may grant the Keystone for assessed offline-equivalent work (logged, audited — see Teacher Guide).
- **Composite pass rule:** stage scores compose into the mission pass-threshold via the engine (default 0.60). Tokens reward process; the Keystone records "met the bar for this region."